THE COMPLEX PROJECTOR SET OF $\mathbb{CP}^2$ HAS BORSUK NUMBER NINE

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ABSTRACT. OpenAI's family-156 preprint shows that the projector image of $\mathbb{RP}^3$ in $\mathbb{R}^9$ cannot be covered by ten sets of smaller diameter. The complex analogue, the projector image X of $\mathbb{CP}^2$ in the eight-dimensional space of trace-one Hermitian 3×3 matrices, is a natural candidate for an eight-dimensional counterexample. We record that it is not one: $b(X) = 9$. The lower bound is the verbatim complex transfer of the preprint's odd-map argument. The upper bound is an explicit cover by nine caps, certified on all of $\mathbb{CP}^2$ by a branch-and-bound computation in outward-rounded interval arithmetic and, independently, in exact integer arithmetic. The result is a negative one. Since $\mathbb{CP}^n$ for $n \geq 3$ lives in dimension ≥ 15, it closes the route of complex projective projector sets for counterexamples in dimension < 9. The mathematics is elementary.

1. STATEMENT

For a unit vector $u \in \mathbb{C}^3$ write $P_u = uu^*$. Let

$$X = \{P_u : \ u \in \mathbb{C}^3,\ |u| = 1\} \subset \{A \in \mathrm{Herm}_3(\mathbb{C}) : \mathrm{tr}\, A = 1\} \cong \mathbb{R}^8,$$

with the Frobenius metric $\|A\|_F^2 = \mathrm{tr}(A^*A)$. Since $\mathrm{tr}(P_uP_v) = |\langle u, v\rangle|^2$,

$$\|P_u - P_v\|_F^2 = 2 - 2|\langle u, v\rangle|^2, \tag{1}$$

so $\mathrm{diam}\, X = \sqrt{2}$, attained exactly at orthogonal pairs. For a bounded set Y of positive diameter, $b(Y)$ is the least number of subsets of diameter $< \mathrm{diam}\, Y$ that cover Y.

Theorem 1. $b(X) = 9$. *Explicitly, there are nine points $p_1, \ldots, p_9 \in \mathbb{C}^3$ (listed in the accompanying file `P9.txt`) such that the nine closed caps*

$$C_i = \left\{P_u \in X : \ |\langle u, p_i\rangle|^2 \ \geq \ \tfrac{21}{40}\,|p_i|^2\right\}$$

cover X and each has diameter at most $\sqrt{399/200} < \sqrt{2}$. Conversely X cannot be covered by eight sets of diameter $< \sqrt{2}$.

Since X spans an 8-dimensional affine space, $9 = 8 + 1$ is exactly Borsuk's bound, so X is not a Borsuk counterexample.

2. LOWER BOUND: $b(X) \geq 9$

This is the complex form of the preprint's strict-gap reduction (`geometry.tex`, `lem:cover-to-map`), of its odd extension (`extension.tex`, `lem:positive-extension`, `lem:signed-extension`) and of the first sentence of its Proposition `thm:label-bound`. The proofs carry over line by line. We indicate the only changes.

Proposition 2. *If X is covered by m sets of diameter $< \sqrt{2}$, then $m \geq 9$.*

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Proof. Step 1 (reduction). As in the preprint (proof of `lem:cover-to-map`), enlarge the sets to relatively open sets U_i of diameter $< \sqrt{2}$ and take a continuous partition of unity on $\mathbb{CP}^2$ subordinate to their preimages. This gives continuous $f = (f_1, \dots, f_m) : \mathbb{CP}^2 \to \Delta^{m-1}$ such that, with $S(x) = \{i : f_i(x) > 0\}$, orthogonal lines $x \perp y$ have $S(x) \cap S(y) = \varnothing$. (If $i \in S(x) \cap S(y)$ then $P_x, P_y \in U_i$ would be at distance $\sqrt{2}$ by (1).) Only continuity is needed below.

Step 2 (extension). Put $H(v) = |v|^2 f([v])$ for $v \neq 0$, $H(0) = 0$; it is continuous because $\|H(v)\|_1 = |v|^2$. For positive semidefinite $Q \in \mathrm{Herm}_3$ let

$$E(Q) = 3 \int_{S(\mathbb{C}^3)} H(Q^{1/2}u)\, d\mu(u),$$

with μ the $U(3)$-invariant probability measure on the unit sphere of $\mathbb{C}^3$. The map E is continuous, since $Q \mapsto Q^{1/2}$ is continuous on the PSD cone. It is positively homogeneous and coordinatewise nonnegative. Since $\int uu^*\, d\mu = \frac{1}{3} I$, it satisfies $\sum_i E(Q)_i = 3 \int \langle Qu, u\rangle d\mu = \operatorname{tr} Q$. As in the preprint, $\{i : E(Q)_i > 0\} = \bigcup_{x \subseteq \operatorname{ran} Q} S(x)$. This uses only that $Q^{1/2}$ maps the sphere onto a set spanning every line of $\operatorname{ran} Q$, and that nonempty open subsets of the sphere have positive measure.

Step 3 (odd map). For $A \in \mathrm{Herm}_3$ write $A = A_+ - A_-$ with $A_\pm \succeq 0$ supported on the orthogonal positive and negative spectral subspaces, and put $F(A) = E(A_+) - E(A_-)$. Every line in $\operatorname{ran} A_+$ is orthogonal to every line in $\operatorname{ran} A_-$. By Step 1 the two vectors therefore have disjoint supports, so $\|F(A)\|_1 = \operatorname{tr} A_+ + \operatorname{tr} A_- = \operatorname{tr} |A|$. Moreover F is continuous and odd ($A_\pm$ are exchanged under $A \mapsto -A$). Hence $A \mapsto F(A)$ restricts to a continuous odd map from $\{A \in \mathrm{Herm}_3 : \operatorname{tr}|A| = 1\} \cong S^8$ (as $\dim_{\mathbb{R}} \mathrm{Herm}_3 = 9$) to $\{y \in \mathbb{R}^m : \|y\|_1 = 1\} \cong S^{m-1}$. By the Borsuk–Ulam theorem, $m - 1 \geq 8$. □

Remark 3 (What is and is not analogous). For real $k \times k$ symmetric matrices the same argument gives $m \geq k(k+1)/2$, which is again Borsuk's bound $d + 1$. The preprint's counterexample comes from excluding the *equality case* $m = N_4 = 10$ on $\mathbb{RP}^3$, by a long degree-theoretic and combinatorial argument (`topology.tex`, `six-labels.tex`, `ten-labels.tex`). Note that $\mathbb{CP}^2$ is the complex analogue of $\mathbb{RP}^2$ ($k = 3$, six labels), where the preprint does not exclude equality. The complex analogue of the $k = 4$ witness is $\mathbb{CP}^3 \subset \mathbb{R}^{15}$, which is irrelevant for the dimension record. The point of Theorem 1 is only that the single complex projector set of dimension below 9 that is not a round sphere, namely X in $\mathbb{R}^8$, is not a counterexample. ($\mathbb{CP}^1$, $\mathbb{HP}^1$ and $\mathbb{OP}^1$ give round spheres S^2, S^4, S^8 in $\mathbb{R}^3, \mathbb{R}^5, \mathbb{R}^9$, for which $b = d + 1$. $\mathbb{HP}^2$ lives in $\mathbb{R}^{14}$.)

3. Upper bound: nine caps

Lemma 4 (Caps)**.** *Let $p, x, y \in \mathbb{C}^3$ be unit vectors with $|\langle x, p\rangle|^2 \geq T$ and $|\langle y, p\rangle|^2 \geq T$, where $T > \frac{1}{2}$. Then $|\langle x, y\rangle| \geq 2T - 1$, and hence $\|P_x - P_y\|_F^2 \leq 2 - 2(2T-1)^2$.*

Proof. Write $x = ap + x'$ and $y = bp + y'$ with $x', y' \perp p$. Then $|a|^2 \geq T$ and $|x'|^2 = 1 - |a|^2 \leq 1 - T$, and likewise for y. Hence $|\langle x, y\rangle| = |a\bar{b} + \langle x', y'\rangle| \geq |a||b| - |x'||y'| \geq T - (1 - T)$. Apply (1). □

With $T = 21/40$ this gives $\operatorname{diam} C_i \leq \sqrt{2 - 2/400} = \sqrt{399/200} \approx 1.41244$. It remains to show that the caps cover X. Because the condition is homogeneous, this is the statement

$$\forall x \in \mathbb{C}^3 \setminus \{0\} : \qquad \max_{1 \leq i \leq 9} h_i(x) > 0, \qquad h_i(x) = |\langle x, p_i\rangle|^2 - T\,|p_i|^2 |x|^2. \tag{2}$$

Proposition 5 (computer-assisted)**.** (2) *holds for the nine points of `P9.txt` with $T = 21/40$, and also with $T = 263/500 = 0.526$.*

Domain. Every point of $\mathbb{CP}^2$ has a representative with $x_j = 1$ and $|x_k| \leq 1$ for $k \neq j$: divide by the coordinate of largest modulus. Then $(\operatorname{Re} x_k, \operatorname{Im} x_k)_{k \neq j} \in [-1, 1]^4$. So it suffices to prove

(2) on three boxes $[-1,1]^4$ (one per chart j). Each h_i is a real quadratic polynomial in the four free coordinates.

Primary certificate (`cert_interval.py`). Branch and bound with 16-way splitting. On each box, Re and Im of $\langle x, p_i\rangle$ are linear forms, enclosed in interval arithmetic; their squares and $|x|^2$ are enclosed termwise. A box is accepted when the lower endpoint of the enclosure of some h_i is > 0. Every IEEE-754 operation is followed by an outward `nextafter` step, so all enclosures are rigorous. The data p_i are exact binary64 numbers, and $T|p_i|^2$ is computed exactly as a rational and then enclosed. For $T = 21/40$: 1 897 683 accepted leaves (minimum half-width 2^{-12}), under a minute on one core.

Independent certificate (`independent/cert_exact.py`, standard library only). We use exact integer arithmetic throughout. For a quadratic H and a box $c \pm r$, it uses the exact expansion $H(c+d) = H(c) + \nabla H(c)\cdot d + d^{\mathsf{T}} Q d$ and the lower bound

$$H(c) - \sum_k |\partial_k H(c)| r_k + \sum_k \min(Q_{kk}, 0) r_k^2 - \sum_{k\neq l} |Q_{kl}| r_k r_l.$$

Boxes are bisected along the widest coordinate. For $T = 21/40$: 16 876 leaves, maximal depth 36, about 4 s.

A symmetric configuration is not enough. For the Hesse SIC (orbit of $(0,1,-1)/\sqrt{2}$ under the Heisenberg group) every overlap of the basis vector e_1 with the nine SIC lines is 0 or $\frac{1}{2}$, so its maximal overlap is exactly $\frac{1}{2}$ and its caps only reach the borderline $T = \frac{1}{2}$, i.e. diameter $\sqrt{2}$. A perturbed, computer-found configuration is therefore used.

Sharpness and control. Numerical minimization gives $\min_x \max_i |\langle \hat{x}, \hat{p}_i\rangle|^2 \approx 0.526055$ for these points. Both certificates succeed at $T = 0.526$. At $T = 0.53$ the primary code stops with an explicit violating point $x = (1,\ 0.67578125 + 0.30859375i,\ -0.26171875 + 0.08984375i)$, whose normalized maximal overlap is $0.52950 < 0.53$. Deleting the ninth point also makes the code fail immediately, as Proposition 2 predicts for any eight caps with $T > \frac{1}{2}$.

Proof of Theorem 1. Proposition 2 gives $b(X) \geq 9$. Proposition 5 and Lemma 4 give nine closed caps that cover X, each of diameter $\leq \sqrt{399/200} < \sqrt{2}$. □

4. Provenance, scope and what is not claimed

The nine points were found by a heuristic max–min search (soft-min objective, L-BFGS-B). The search is not part of the proof. The certificate concerns only the fixed exact numbers in `P9.txt`.

- We do not determine the best covering threshold $\max_{p_1..p_9} \min_x \max_i |\langle \hat{x}, \hat{p}_i\rangle|^2$ (it is at least 0.526), nor whether symmetric configurations (SIC/MUB/Clifford-equivariant) work. An earlier exploratory computation suggested that Clifford-equivariant nine-covers have margin at most 0.005. That computation is not used here.
- Nothing about dimensions ≤ 8 in general follows. The note shows only that this particular candidate set does not give a Borsuk counterexample in $\mathbb{R}^8$.
- The lower bound is a direct transfer of the preprint's argument. The upper bound is an elementary covering-radius computation. We do not claim new methods.
- There is no Lean formalization.

References

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Developed with AI assistance (Anthropic Claude, Opus 5.5) under [AUTHOR NAME]'s direction; independent adversarial reviews were run by separate AI sessions and are not human peer review.